

Static Gesture Classification — Run Report

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Data paths

Training data path 1	/home/jestin/ThesisRepo/ML/NewTrainingData/Static · 425 trials
Total training paths	1 · 425 trials total
External test root	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Static
Report output dir	/home/jestin/ThesisRepo/ML/saved_reports

Sensor selection

Hands	left, right
Segments	palm_mid, thumb, index, middle, ring, pinky, wrist
Modalities	YPR, Accel, Flex
Sensor columns	176 channels
Classes	7: Double_Closed_Fist, Double_Nothing, Double_Okay, Double_Open_Palm, Double_Phone, Double_Pistol, Double_Spiderman

Preprocessing, features & split

Resample steps	90
Butterworth	on · cutoff 5.0 Hz · order 4 · fs 30.0 Hz
Normalisation	minmax
Feature mode	stats → feature dim 1408
Test size · seed · CV folds	0.2 · random_state=42 · 10-fold stratified
Sample counts	train (post-aug)=680 · test=85

Augmentation (training only)

Enabled	yes
Copies per train trial	1
Jitter	off · sigma=0.01
Amplitude scale	on · range=(0.8, 1.2)
Time warp	off · range=(0.9, 1.1)

Classifiers — cross-validation & hold-out test

Algorithm	CV mean (10-fold)	CV std	Hold-out test acc
SVM (RBF)	1.0000	0.0000	1.0000
SVM (Linear)	1.0000	0.0000	1.0000
Random Forest	1.0000	0.0000	1.0000
Logistic Regression	1.0000	0.0000	1.0000
Voting Soft (SVM+RF+KNN+LogReg)	1.0000	0.0000	1.0000
Stacking (SVM+RF+KNN+LogReg)	0.9971	0.0088	1.0000
KNN	0.9868	0.0103	0.9882
LDA	0.9471	0.0310	0.5529

Per-class hold-out F1

Class	SVM (RBF)	SVM (Linear)	Random Forest	Logistic Regression	Voting (soft)	Stacking	KNN	LDA	Support
Double_Closed_Fist	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.600	13
Double_Nothing	1.000	1.000	1.000	1.000	1.000	1.000	0.957	0.316	12
Double_Okay	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.636	12
Double_Open_Palm	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.571	12
Double_Phone	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.364	12
Double_Pistol	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.467	12
Double_Spiderman	1.000	1.000	1.000	1.000	1.000	1.000	0.960	0.846	12
macro avg	1.000	1.000	1.000	1.000	1.000	1.000	0.988	0.543	85

External test (NewTestData)

Algorithm	External test acc
Random Forest	0.9167
Stacking (SVM+RF+KNN+LogReg)	0.9167
SVM (RBF)	0.8889
KNN	0.8889
Voting Soft (SVM+RF+KNN+LogReg)	0.8889
SVM (Linear)	0.8611
Logistic Regression	0.8611
LDA	0.3611

Per-class external accuracy

[illegible]